Effect of Seismic Uplift Pressure on the Behavior of Concrete Gravity Dams with a Penetrated Crack

O. A. Pekau[1] and Xueye Zhu[2]

Abstract: Proper consideration of the uplift pressure at the base of a concrete gravity dam is of great importance in practical engineering, since it is crucial to the safety of the dam, specifically for a cracked dam under seismic conditions. However, constant uplift pressure, which is suitable for the static case only, was adopted in almost all the seismic analyses of cracked concrete gravity dams. To adequately estimate the seismic behavior of cracked concrete gravity dams, a seismic uplift pressure model is proposed for a penetrated crack. In this model, the amount and the distribution of the uplift pressure along the assumed rigid crack walls are determined by the earthquake acceleration, the water heads, the aperture of the crack, and the opening/closing velocity. Application of the model to a typical concrete gravity dam with a penetrated crack at the base reveals that the seismic behavior of the dam is markedly affected by the seismic uplift pressure. In general, the residual downstream sliding is considerably enlarged compared to that of constant uplift pressure. Computations show that the seismic uplift pressure can be several times higher than the constant one, increasing the dynamic instability of the cracked dam. It is also revealed that the dynamic water flow plays the role of a wedge while the upper mouth of the crack is closing. When the dam rocks back to upstream, the uplift pressure increases until it is so high that the pivot at the toe is raised up and the whole dam loses its contact. Then the resultant uplift pressure remains constant until the dam is inclined to the upstream. During this period of time, the cracked dam is normally drifting towards the downstream due to the hydro pressure.

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Introduction

Seismic behavior of cracked concrete gravity dams has been studied in the past decades, and the main efforts were focused on the effects of crack propagating in the concrete body of the dam (Bhattacharjee and Léger 1993; Ayari and Saouma 1990; El-Aidi and Hall 1989). In these cases, the addressed features are the fracture failure of the material and its influence on the seismic response of the dam, without much concern for the safety of the damaged dam in sliding and overturning. Nevertheless, once the crack develops to such an extent that it has penetrated through the dam, no matter whether it locates at the base or at a height, the safety consideration of the part above the crack is obviously demanded.

Chopra and Zhang (1991) explored the earthquake induced base sliding of gravity dams with a one-degree-of-freedom model, where a constant cutoff triangular distribution of uplift pressure was adopted. Chávez and Fenves (1995) assumed a constant triangular uplift pressure in their study of the earthquake response of gravity dams with a crack at the base by use of the finite-element method. Similar to Chopra and Zhang (1991), Chávez and Fenves (1995) ignored the small rocking movement at the base, since it was considered insignificant compared with sliding. Léger and Katsouli (1989) and Fronteddu et al. (1998) modeled the base interface and lift joints with gap-friction elements, whereas the uplift pressure was still constant, the same as in the smeared-crack approach (El-Aidi and Hall 1989).

In reality, the uplift pressure will change in magnitude and in distribution during an earthquake along the crack that is either propagating or stabilized. This is due to both dynamic hydro pressure at the crack mouth initiated directly by the earthquake and the opening-closing movement of the crack walls accompanying the earthquake induced rocking movement of the dam. To consider the transient nature of the uplift pressure, Hall (1998) assumed that the pressure at the mouth of the crack varies in accordance with the hydro dynamic pressure due to the dam-reservoir interaction, and the uplift pressure maintains linear variation in the crack. Zhang and Ohmachi (1998) proposed a similar seismic uplift pressure model for seismic crack analysis, neglecting the effect of the crack wall motion on the transient pressure. Slowik and Saouma (2000) developed a theoretical model for simulation analysis of crack propagation in concrete at the initial phase of cracking during an earthquake. Their tests indicated that the hydro pressure in the crack can be as high as three times that of the initial static one during sudden crack closure. With the assumptions that one-dimensional laminar water flows in a water filled existing rectangular crack of constant length and that the crack wall motion is small in amplitude, Tinawi and Guizani (1994) formulated an analytical equation for the seismic uplift pressure, and then applied it in a case study of a typical gravity dam subject to earthquake loading, showing that the maximum pressure can be as high as two times the static

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uplift pressure. Javanmardi et al. (2005b) conducted experiments of transient water pressure for a propagating crack in concrete, and proposed a theoretical model, considering the motion of the crack wall as rigid and assuming one-dimensional laminar and turbulent water flows in the saturated region of the crack. Both the experiments and the theoretical model indicated that the seismic uplift pressure can be several times that of the static one. As an application of this model, Javanmardi et al. (2005a) studied the crack propagation at the base of a 90 m high typical concrete gravity dam to evaluate the safety in sliding under seismic conditions.

In contrast to the above studies of crack propagation where the structure is still intact, concrete gravity dams with a penetrated crack, which attracted much attention (Pekau and Cui 2004; Pekau and Zhu 2006), are no longer a single structure. In this case, the concrete gravity dam is separated into two parts by the crack, and the upper part above the penetrated crack behaves like a free standing block resting on a flexible frictional surface. With constant triangular uplift pressure assumption, it is shown that the residual sliding can be relatively large under strong earthquake conditions. Of greater interest is that the upper part may even experience drifting mode of motion, and small rocking cannot be ignored since it is correlated with sliding. This indicates that the safety of this kind of fractured gravity dams may be more crucial, if seismic uplift pressure which can be higher than the static one is considered.

This paper presents a theoretical model, inspired by the research of Javanmardi et al. (2005b), describing the seismic uplift pressure in a penetrated crack to improve the investigation of the seismic behavior of concrete gravity dams with a penetrated crack. After the model is described, a 65 m high typical concrete gravity dam is studied to expose the importance of the seismic uplift pressure to reveal more practically the real features of the seismic behavior of cracked concrete gravity dams.

Seismic Uplift Pressure Model for Penetrated Crack

Cracks in concrete gravity dams are usually initiated at the lift joints, the base of the dam, and the locations where the cross section changes abruptly. Due to the properties of concrete, the profile of a crack is irregular in engineering practice. To simplify the analysis, however, it is assumed that the penetrated crack in this study is straight and flat. Although the local deformation along the crack wall is considerable when the crack is closing, the crack wall is assumed to be rigid when the geometry of the cracked wall is considered.

For a penetrated crack, both ends of the crack may experience opening and closing due to overall rocking of the dam in both directions. When the dam rocks toward the downstream with the pivot at its toe, the upstream mouth opens, and the transient uplift pressure decreases, which is similar to the case of opening a propagating crack. After the rocking reaches its peak, the dam will rock back with the same pivot, and the transient uplift pressure will increase. In this case, the water pressure at the downstream mouth is known since the downstream water head is given, which is different from the case of a propagating crack where, instead, the water flow is zero at the tip of the crack and the pressure there can be very high. As for the case of rocking with pivot at the heel of the dam, the water pressure is, under the assumption of incompressible water, obviously negligible since water flow from the upstream is blocked. However, there exists the possibility that both ends are opening and/or closing

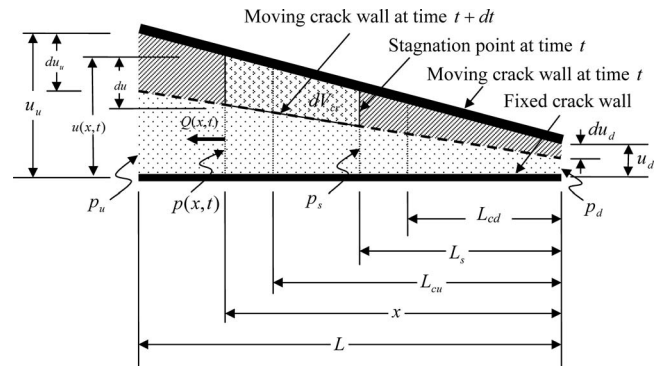

Fig. 1. Penetrated crack in closing

in the penetrated crack, as the motion mode of drifting was observed in previous studies (Pekau and Cui 2004; Pekau and Zhu 2006). Therefore the seismic uplift pressure model can be illustrated in Fig. 1 for a penetrated crack, where L=length of the crack; L_s=position of the stagnation point of the water flow; L_{cu} and L_{cd}=positions where the water flow changes from laminar to turbulent in upstream and downstream directions, respectively; u_u and u_d=openings at the upstream mouth and downstream mouth, respectively; du_u and du_d=differentials of u_u and u_d, respectively; $Q(x,t)$=water flow; p_u and p_d=known water pressures at the upstream and the downstream mouths, respectively; and p_s=unknown water pressure at the stagnation point.

Under the rigid crack wall assumption, the aperture of the crack can be expressed in terms of the openings at the mouths as follows:

$$u(x,t) = \frac{x}{L}u_u + \left(1 - \frac{x}{L}\right)u_d \tag{1}$$

With Stagnation Point

For a lengthy crack there may exist a stagnation point, where the water does not flow as shown in Fig. 1. Assuming that the crack walls are impermeable and the water is incompressible, the water quantity dV_{cr} flowing through the section at x during the time dt will be

$$dV_{\mathrm{cr}} = \frac{x - L_s}{2}\left[\frac{x + L_s}{L}du_u + \left(2 - \frac{x + L_s}{L}\right)du_d\right] \tag{2}$$

Therefore the water flow at section x will be

$$Q(x,t) = \frac{dV_{\mathrm{cr}}}{dt} = \frac{x - L_s}{2}\left[\frac{x + L_s}{L}\dot{u}_u + \left(2 - \frac{x + L_s}{L}\right)\dot{u}_d\right] \tag{3}$$

where $\dot{u}_u = du_u/dt$ and $\dot{u}_d = du_d/dt$=closing velocities at the upstream mouth and the downstream mouth, respectively, $Q(x,t) > 0$ for the upstream section ($x > L_s$), and $Q(x,t) < 0$ for the downstream section ($x < L_s$).

It is obvious that the water flow near the stagnation point is low and laminar in form. Apart from the stagnation point, the water quantity dV_{cr} increases and the water flow is possibly turbulent. In the upstream section, assume that the water flow changes from laminar to turbulent at the position L_{cu} determined by the computed Reynold's number (Wittke 1990)

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$$\mathsf{R} = \frac{2\rho Q(x,t)}{\nu} \tag{4}$$

where ρ=density of water, and ν=dynamic viscosity of water. When $\mathsf{R} \leqslant 2,300$ the water flow is laminar and when $\mathsf{R} > 2,300$ it is turbulent.

The pressure in the upstream section is

$$p(x,t) = \begin{cases} p_s + \int_{L_s}^{x} dp_l(x,t) & L_s < x \leqslant L_{cu} \\ p_s + \int_{L_s}^{L_{cu}} dp_l(x,t) + \int_{L_{cu}}^{x} dp_t(x,t) & L_{cu} < x \leqslant L \end{cases} \tag{5}$$

where the subscripts l and t denote the laminar and turbulent flows, respectively; and the pressure gradients for laminar and turbulent flow can be, respectively, expressed as (Wittke 1990)

$$dp_l(x,t) = 12\rho\nu\left[1 + 8.8\left(\frac{k}{2u(x,t)}\right)^{1.5}\right]\frac{Q(x,t)}{u^3(x,t)}dx \quad (\mathsf{R} \leqslant 2{,}300) \tag{6}$$

$$dp_t(x,t) = \frac{\rho}{16\left[\log\frac{1.9}{k/2u(x,t)}\right]^2}\frac{Q^2(x,t)}{u^3(x,t)}dx \quad (\mathsf{R} > 2{,}300) \tag{7}$$

and $k/2u(x,t)$=relative crack roughness, which is taken as 0.5 in this study for a tensile crack (Javanmardi et al. 2005b).

Eqs. (6) and (7) show that the pressure gradient is proportional to $Q(x,t)$ and $Q^2(x,t)$ for laminar and turbulent flows, respectively. It is also noted that the pressure gradient is inversely proportional to the crack aperture $u^3(x,t)$. This indicates that with the same closing velocity, higher pressure will be created in the case of thinner aperture. Specifically, the pressure gradient could be infinite at the instant of impact when the crack is being closed completely. If this occurs, the computation error of up-flow will appear. It is pointed out that a crack can never be closed perfectly due to the existence of crushed concrete particles (Javanmardi et al. 2005a). Therefore, it is reasonable to assume that there exists a nonzero residual aperture u_r for the penetrated crack, i.e.

$$u(x,t) = u_r + u_m(x,t) \tag{8}$$

where u_m=mechanical opening of the crack in accordance with the displacement field of the system.

Similarly, for the downstream section of the crack ($x < L_s$), the water pressure is given by

$$p(x,t) = \begin{cases} p_s - \int_{L_s}^{x} dp_l(x,t) & (L_{cd} \leqslant x < L_s) \\ p_s - \int_{L_s}^{L_{cd}} dp_l(x,t) - \int_{L_{cd}}^{x} dp_t(x,t) & (0 < x \leqslant L_{cd}) \end{cases} \tag{9}$$

For a given status of closing there are two unknowns, the location of the stagnation L_s and the corresponding pressure p_s, in the uplift pressure model as expressed in Eqs. (5) and (9). Due to known pressures p_u and p_d at the two mouths of the crack, respectively, Eqs. (5) and (9) lead to two equations for L_s and p_s

$$p_u = p_s + \int_{L_s}^{L_{cu}} dp_l(x,t) + \int_{L_{cu}}^{L} dp_t(x,t) \tag{10}$$

$$p_d = p_s - \int_{0}^{L_{cd}} dp_t(x,t) - \int_{L_{cd}}^{L_s} dp_l(x,t) \tag{11}$$

Therefore an integral equation of unknown L_s is obtained from Eqs. (10) and (11)

$$p_u - p_d = -\int_{0}^{L_{cd}} dp_t(x,t) - \int_{L_{cd}}^{L_s} dp_l(x,t) + \int_{L_s}^{L_{cu}} dp_l(x,t) + \int_{L_{cu}}^{L} dp_t(x,t) \tag{12}$$

A natural method to solve this complicated equation is by iteration. Once L_s is determined, the pressure p_s at the stagnation point can be obtained from either Eq. (10) or (11). Then the uplift pressure at time t will be determined with Eqs. (5) and (9).

Without Stagnation Point in Fully Saturated Crack

However, Eq. (12) may not have a realistic solution to L_s requiring $0 \leqslant L_s \leqslant L$. This is because the closing velocities may not be large enough or one or both mouths of the crack may be opening instead of closing. In this case, the water flow can be written as follows:

$$Q(x,t) = \frac{x}{2}\left[\frac{x}{L}\dot{u}_u + \left(2 - \frac{x}{L}\right)\dot{u}_d\right] + Q_0 \tag{13}$$

where Q_0=unknown constant water flow.

Similarly to Eq. (12), the equation for unknown Q_0 will be given by

$$p_u - p_d = \int_{0}^{L} dp(x,t) \tag{14}$$

where the pressure gradient dp is dp_l in Eq. (6) or dp_t in Eq. (7), depending on the computed Reynold's number. Once the water flow Q_0 is found through iteration, the distribution of the uplift will be

$$p(x,t) = p_d + \int_{0}^{x} dp(x,t) \tag{15}$$

Without Stagnation Point in Partially Saturated Crack

It is should be noted that the water flow Q_0 must be positive, meaning that the water flow is downstream at the downstream mouth. If the assumption of full saturation is held, however, this will be violated when one or both ends of the crack open with velocities that exceed certain limits. In reality, the water has no time to fill the entire crack in this situation, leaving an unsaturated segment near the downstream mouth. Therefore, it is required to determine the actual length L_a of saturation beyond which the uplift pressure is p_d. In this case, two unknowns, L_a and Q_0, appear. Now the problem becomes one to find a maximum L_a ($0 < L_a < L$) under the condition of $Q_0 > 0$ to satisfy the equation

$$p_u - p_d = \int_{L_a}^{L} dp(x,t) \tag{16}$$

where the pressure gradient will be calculated according to Eqs. (6) and (7), and the corresponding water flow is changed to

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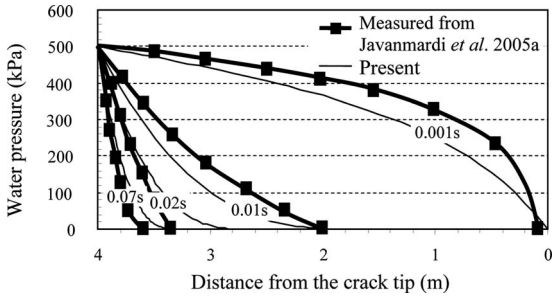

Fig. 2. Comparison of uplift pressure in 4 m crack when opening

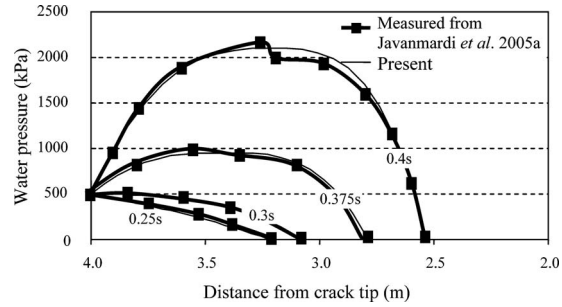

Fig. 3. Comparison of uplift pressure in 4 m crack when closing

$$Q(x,t) = \frac{x - L_a}{2}\left[\left(\frac{x}{L_a} - 1\right)\dot{u}_u + \left(3 - \frac{x}{L_a}\right)\dot{u}_d\right] + Q_0 \quad (L_a \leqslant x \leqslant L) \tag{17}$$

This is a double-loop iteration procedure. With L_a, the corresponding Q_0 in Eq. (16) can be found through iteration as in Eq. (14) with $x=0$ replaced by $x=L_a$. Then L_a will be updated according to the obtained Q_0 until Q_0 is positive. Therefore, the uplift pressure in this case is given by

$$p(x,t) = \begin{cases} p_d & (x < L_a) \\ p_d + \displaystyle\int_{L_a}^{x} dp & (L_a \leqslant x \leqslant L) \end{cases} \tag{18}$$

Verification and Discussion of Model

Since no experimental data are available for the uplift pressure in penetrated cracks, the computed uplift pressure in Javanmardi et al. (2005b) for an existing 4 m long nonpenetrated crack is adopted to verify the proposed model. The crack had assumed constant pressure of 500 kPa at the crack mouth with an initial opening of 0.5 mm. The motion of the crack wall was harmonic with amplitude of 1.0 mm and frequency of 2 Hz, namely

$$u_u = 1.0 + 0.5\sin(1.5 + 4t)\pi \text{ mm} \tag{19}$$

In the first 0.25 s, the crack mouth is open, and the saturation length of the crack decreases. Since the initial water flows and water pressures along the crack in the two models are different, it is not expected to have the same computed pressure for the two cases. However, once the crack wall starts to open and cavitation occurs, the water pressure within the unsaturated portion in both models becomes the same, which is zero. This makes the water pressure comparable for the two models.

Fig. 2 illustrates the uplift pressure computed in Javanmardi et al. (2005b) and corresponding results presented with residual aperture of 0.3 mm, while the crack mouth is opening. The differences are obviously caused by the difference in initial pressure between the two cases.

After the crack mouth opens to its maximum position at $t=0.25$ s, the saturation length increases as the crack is closing. Since there is no water pressure within the unsaturated portion, the same saturated length in Javanmardi et al. (2005b) is adopted here for the purpose of verification. Fig. 3 shows the computed uplift pressure at times $t=0.25$, 0.30, 0.375, and 0.40 s from the two models. Afterwards, the water pressure at the crack tip builds up very rapidly in Javanmardi et al. (2005b), whereas it is kept at zero at the downstream mouth of the crack in the proposed model. Therefore there is no compatibility for the two models after $t=0.40$ s.

From Figs. 2 and 3, it is seen that the proposed uplift model is consistent with that in Javanmardi et al. (2005b) when the pressures at both *ends* of the crack are the same. Thus the proposed model is suitable for a penetrated crack. It is also noted that the dynamic water pressure is very sensitive to the velocity of the crack wall, for both cases of opening and closing. For example, the saturation length reduces to only 0.6 m in the proposed model when the opening velocity at the crack mouth is just 4.84 mm/s at $t=0.07$ s shown in Fig. 2, losing almost all the uplift pressure compared with the constant case. On the contrary, the pressure is above 2,000 kPa, four times the water pressure of 500 kPa at the crack mouth, when the closing velocity is only 5.98 mm/s at $t=0.4$ s shown in Fig. 3. This phenomenon of transient high uplift pressure observed in the experiments (Javanmardi et al. 2005b) is consistent with the observation of Slowik and Saouma (2000) that when the crack closes rapidly the water can be trapped in the crack, acting as a wedge and even causing the tensile stresses in the ligament. This suggests the occurrence of transient overall uplift of the dam with a penetrated crack under strong earthquake shocks.

Taking the typical penetrated crack shown in Fig. 4 as an example, the uplift pressure distributions for constant and dynamic cases are illustrated in Fig. 5. It is seen that the 0.10 m/s closing velocity at the upstream mouth will cause eight times the pressure at the mouth. The corresponding resultant uplift force changes from the value of 20.01 MN/m for constant triangular uplift pressure to 196.21 MN/m. Nevertheless, if the 0.10 m/s velocity is for opening, the saturation length shrinks to less than 10 m, with resultant uplift force of 1.69 MN/m, just 8.45% of the constant value. This violent change of uplift pressure owing to the velocity of the crack wall indicates that the seismic behavior of cracked concrete gravity dams will be markedly affected by the seismic uplift pressure in the penetrated crack.

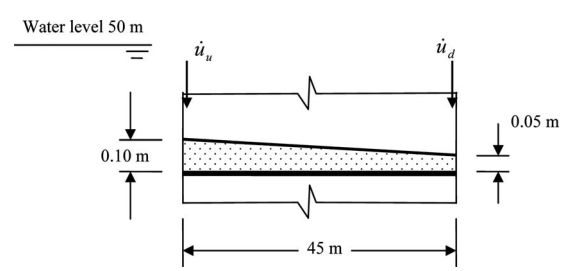

Fig. 4. Forty-five m penetrated crack with 50 m water

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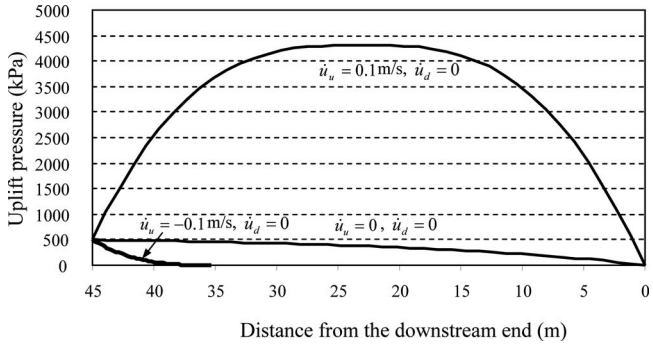

Fig. 5. Seismic and constant pressures of 45 m penetrated crack with 50 m water

Effect of Seismic Uplift Pressure on Rocking of Gravity Dam with Penetrated Crack

When the seismic uplift pressure is considered, the net weight of the dam and the rocking moment vary according to the earthquake acceleration and the seismic response of the dam. Therefore, the motion mode of the dam is possibly different from that with constant uplift pressure. To properly understand the seismic behavior, it is worthwhile to study the forces applied on the dam during the seismic procedure.

Considering the case when the dam is rocking downstream with the pivot at the toe O_R as shown in Fig. 6, the dynamic equilibrium equations in rotation and in the vertical direction with the assumption that the rotation is relatively small, are given by

$$I_R\ddot{\theta} - M_R = 0 \tag{20}$$

$$P_V + N - W = m_D\ddot{y} \tag{21}$$

where I_R=inertia moment of rotation, including the dynamic effect of water in the horizontal direction (Westergaard 1931), with respect to O_R; m_D and W=, respectively, mass and the weight of the dam; P_V=resultant of the seismic uplift pressure; N=normal contact force acting at O_R where the corresponding friction force is F; θ=angle of rotation; y=vertical displacement

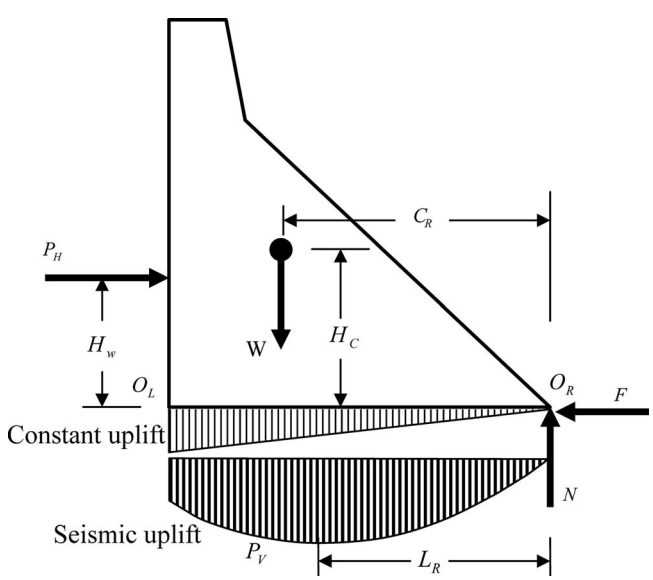

Fig. 6. Forces on cracked gravity dam

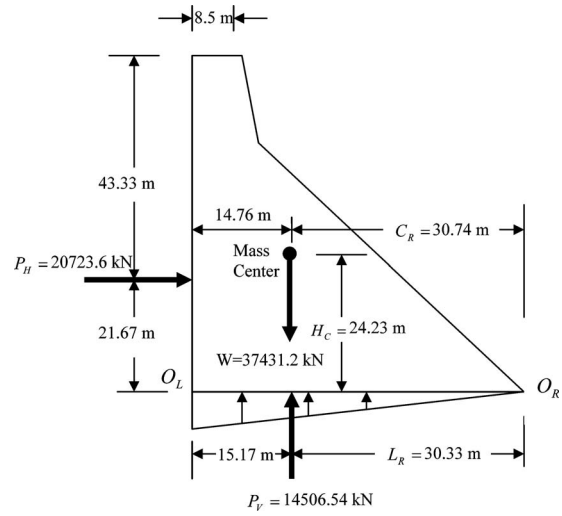

Fig. 7. Typical gravity dam with penetrated crack at base

of the mass center; and M_R=seismic moment with respect to O_R given by

$$M_R = WC_R - P_H H_W - P_V L_R \tag{22}$$

where C_R=distance from the mass center to the toe of the dam; P_H=resultant of the static horizontal hydro pressure; H_W=level of P_H; and L_R=distance from the resultant uplift pressure to the toe.

Since the vertical displacement is related to the rotation by

$$\ddot{y} = -C_R\ddot{\theta} \tag{23}$$

Eqs. (20)–(23) lead

$$N = \left(1 - \frac{m_D C_R^2}{I_R}\right)W + \frac{m_D C_R H_W}{I_R}P_H - \left(1 - \frac{m_D C_R L_R}{I_R}\right)P_V \tag{24}$$

It is noted that the normal contact force N varies with the uplift force P_V and its distance L_R to the pivot. As P_V increases while the crack is closing, L_R decreases accordingly as noted in Fig. 5. Both the increase in P_V and the decrease in L_R will reduce the normal contact force N.

Assuming that the typical gravity dam in Fig. 7 is resuming its position after rocking downstream with pivot at the toe, the closing velocity at the upstream mouth of the crack $\dot{u}_u$ is the only parameter to determine the distribution of the uplift pressure for a specified opening u_u. Fig. 8 illustrates the changes of the resultant uplift force in the crack with 50 mm as the residual aperture for

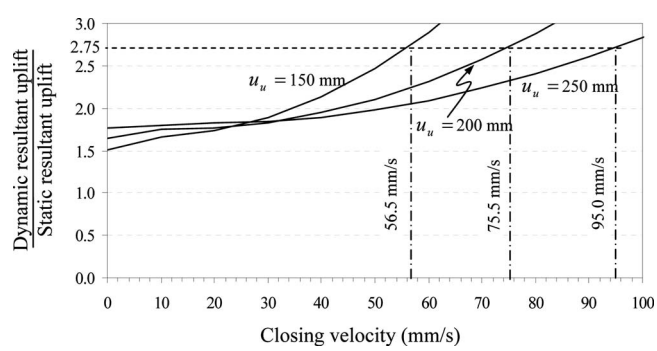

Fig. 8. Resultant uplift pressure in crack of typical gravity dam when crack is closing

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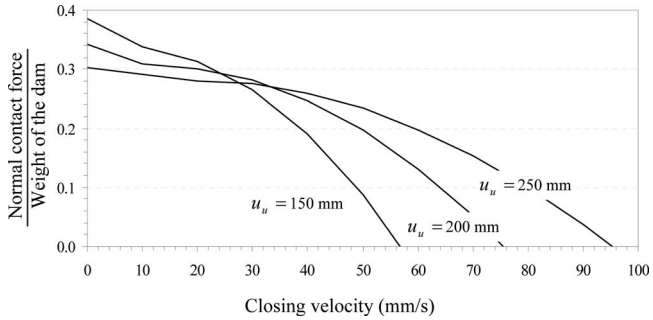

Fig. 9. Normal contact force at toe of typical dam when crack is closing

three openings, 0.25, 0.20, and 0.15 m, versus the closing velocity $\dot{u}_u$. The corresponding variations of normal contact force are shown in Fig. 9.

It is shown in Figs. 8 and 9 that the resultant uplift force increases as the closing velocity increases, lessening the normal contact force. For the openings of 0.15, 0.20, and 0.25 m (the rocking angles are just 0.0033, 0.0044, and 0.0055 rad, respectively), even if the closing velocities are less than 50, 70, and 90 mm/s, respectively, the resultant uplift forces can be above two times that of the constant uplift pressure, and the corresponding normal contact force becomes less than 0.1 times the weight of the dam. This indicates that while the crack is closing, the gravity dam is very likely to initiate or to maintain sliding under the condition of the hydro pressure and the earthquake.

Typically, when the closing velocity reaches such a limit that the pure rocking motion cannot be maintained because the normal contact force in Eq. (24) becomes negative, the dam will lose its contact at the pivot. In this situation, the water flow in the closing crack behaves like a wedge to uplift the pivot while the dam keeps rocking without contact. From Eq. (24), the required uplift force $\bar{P}_V$ is given by

$$\bar{P}_V = \frac{(I_R - m_D C_R^2)W + m_D C_R H_W P_H}{I_R - m_D C_R L_R} \tag{25}$$

Conservatively, assuming the distance L_R=30.33 mm of the constant triangular distribution, the maximum uplift force to lift the pivot is obtained as

$$\bar{P}_V = 2.97 P_{VS} \tag{26}$$

where P_{VS}=resultant uplift force of the constant triangular distribution. $\bar{P}_V$ can be considered as the upper limit of the seismic uplift force to maintain the rocking mode of motion for the typical dam.

It is found that the estimation in Eq. (26) is very close to that obtained from Figs. 8 and 9. According to Fig. 9, the normal contact force will be 0 when the closing velocities reach 56.6, 75.5, and 95.0 mm/s for openings u_u=0.15, 0.20, and 0.25 m, respectively. The corresponding uplift force $\tilde{P}_V \approx 2.75 P_{VS}$ as illustrated in Fig. 8. The difference from $\bar{P}_V$ is obviously due to the fact that the variation of L_R is considered in the computation. Considering that the rocking angle and the closing velocity in Figs. 8 and 9 are relatively small, it is quite convincing to observe the motion mode of rock and drift for a gravity dam with a penetrated crack at the base where larger opening and closing velocities are expected under seismic conditions.

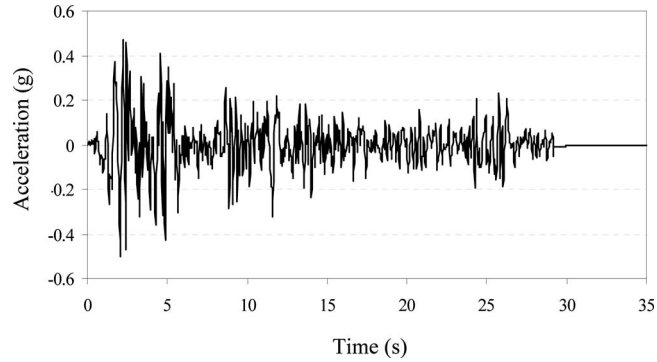

Fig. 10. El-Centro earthquake 1940 NS with peak acceleration scaled to 0.5g

Effect of Seismic Uplift Pressure on Behavior of Cracked Gravity Dams

To reveal the effects of the seismic uplift pressure on the behavior of cracked gravity dams, the typical dam with a penetrated crack at the base as shown in Fig. 7 is studied. The effective coefficient of friction is 1.2, considering the basic Mohr–Coulomb friction and the aggregate interlock. The El-Centro 1940 NS earthquake with peak acceleration scaled to 0.5g (as shown in Fig. 10) and 0.3g are separately imposed at the base of the dam. The density of the dam concrete is 2,500 kg/m^3. Three levels of reservoir water, 50.0, 57.5, and 61.25 m, are considered. Fifty and 75 mm are separately taken as the residual aperture of the crack. Dampers equivalent to the coefficient of restitution of 0.85 are used along the crack to simulate the nonlinearity during the process of impact (Pekau and Zhu 2006).

Residual Downstream Sliding

Computations were carried out by the finite-element method with the incremental displacement constraint equations (IDCE) model (Pekau and Zhu 2006) to deal with the dynamic contact problem. The residual downstream sliding displacements for the typical cracked gravity dam with peak acceleration of 0.3 g and 0.5 g are listed in Table 1, whereas the time histories of the horizontal

Table 1. Residual Downstream Sliding (m) of Typical Dam (Numbers in Parentheses Are Times of Corresponding Values for Constant Uplift Pressure)

Peak acceleration (g)	Water level (m)	Residual aperture 50 mm	Residual aperture 75 mm	Constant uplift case
0.3	50.00	0.006 (0.98)	0.003 (0.54)	0.006
	57.50	0.068 (3.91)	0.026 (1.51)	0.017
	61.25	0.553 (17.74)	0.080 (2.55)	0.031
0.5	50.00	0.038 (1.50)	0.021 (0.84)	0.026
	57.50	0.188 (2.20)	0.116 (1.36)	0.086
	61.25	0.691 (3.89)	0.277 (1.56)	0.178

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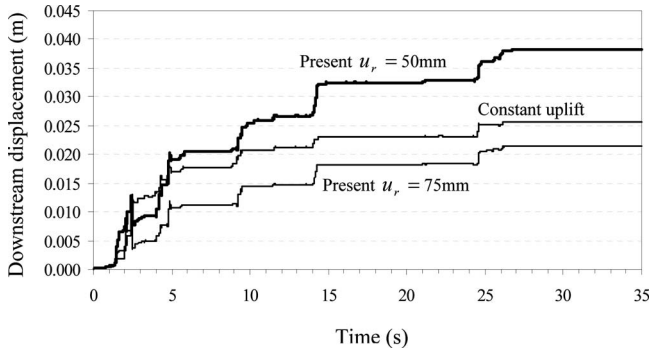

Fig. 11. Downstream residual sliding of typical dam with water level of 50 m under El-Centro 1940 NS 0.5*g*

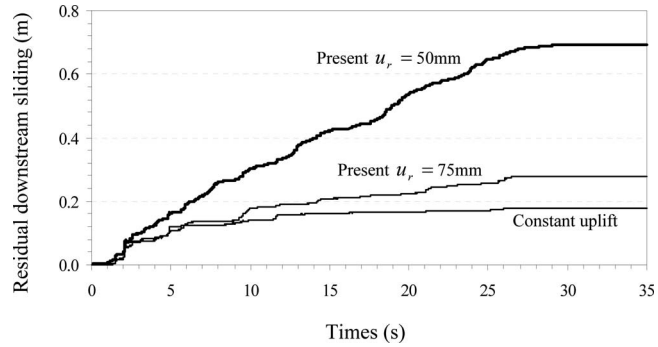

Fig. 13. Downstream residual sliding of typical dam with water level of 61.25 m under El-Centro 1940 NS 0.5*g*

displacement for the three water levels under the El-Centro earthquake with peak acceleration scaled to 0.5 *g* are illustrated in Figs. 11–13.

It is noted from Table 1 that compared with the results for constant uplift pressure, the enlargement in sliding due to seismic uplift pressure is very significant. For peak acceleration of 0.3 *g* and water level at 61.25 m, the residual sliding displacements are 0.553 and 0.080 m for apertures of 50 and of 75 mm, respectively, which are 17.71 and 2.55 times the value of 0.031 m for constant uplift pressure. For peak acceleration 0.5 *g*, the corresponding sliding displacements are 3.89 and 1.60 times, respectively, the value of 0.178 m for constant uplift pressure. These increases are proportional to the water level, but inversely proportional to the residual aperture of the crack. This implies that the seismic uplift pressure makes the residual sliding much larger for higher water level than lower water level, and the smaller the aperture, which means less damage along the crack, the larger the effects of the seismic uplift pressure. This is consistent with the tests (Javanmardi et al. 2005b) stating that the cracks with small residual openings may be more critical than the cracks with greater residual openings.

Comparing the results between the two peak accelerations in Table 1, it is found that the less strong the earthquake imposed, the more marked the increase in sliding. This implies that under minor earthquake conditions a cracked gravity dam that has predicted negligible sliding with constant uplift pressure may actually experience considerable sliding when seismic uplift pressure is considered.

Resultant Uplift Force

In Table 1 it is seen, however, that the sliding of 0.021 m for peak acceleration of 0.5 *g*, water level of 50 m, and residual aperture of 75 mm is less than 0.026 m for constant uplift pressure. This can be explained in Fig. 11 where the dam experienced upstream sliding between 2.0 and 2.5 s when seismic uplift pressure is considered. This implies that the cracked dam may lose its contact on the crack due to seismic uplift pressure on certain occasions as mentioned earlier. The normal contact force for the case shown in Fig. 14 confirms losing contact between t_1=2.14 s and t_2=2.17 s and t_3=2.47 s and t_4=2.51 s.

At t_1=2.14 s, the mode of motion is just switching from rocking with respect to the toe to drifting. At this moment, the computations show that the upper mouth opening is 1.99 mm with closing velocity of 42.20 mm/s, and the lower mouth opening is 0.00 mm with opening velocity of 3.218 mm/s. Therefore, the corresponding resultant uplift force at this moment, according to the proposed uplift model, can be determined to be 3.09 times the resultant of constant uplift pressure with triangular distribution. With the computed apertures and velocities of both ends of the crack, the resultant uplift force around the moment t_1 can be similarly found, of which between 2.13 and 2.16 s is illustrated in Fig. 15. It is seen that the resultant of the seismic uplift pressure increases when the upper mouth of the crack is closing until it is high enough to lift the pivot at the toe of the dam. Then it remains roughly constant until the dam rocks to such a position that the dam is inclined to the upstream when the uplift force drops sud-

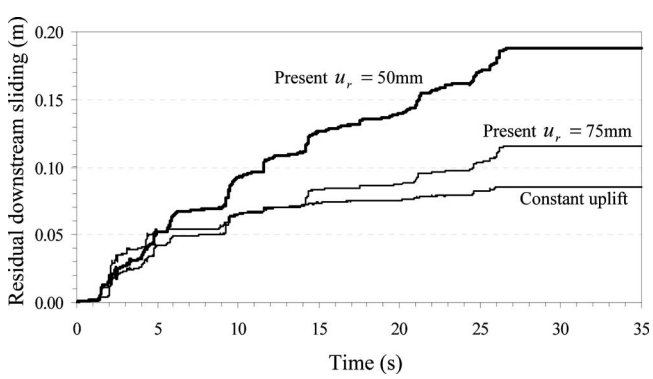

Fig. 12. Downstream residual sliding of typical dam with water level of 57.5 m under El-Centro 1940 NS 0.5*g*

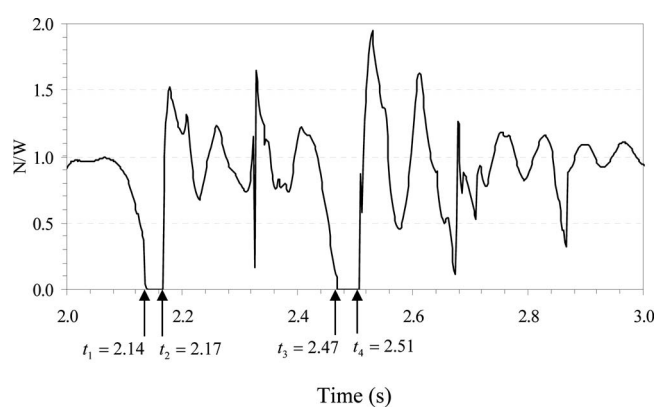

Fig. 14. Ratio of normal contact force to weight of typical dam for 0.5 *g*, 50 m water, and 75 mm residual aperture

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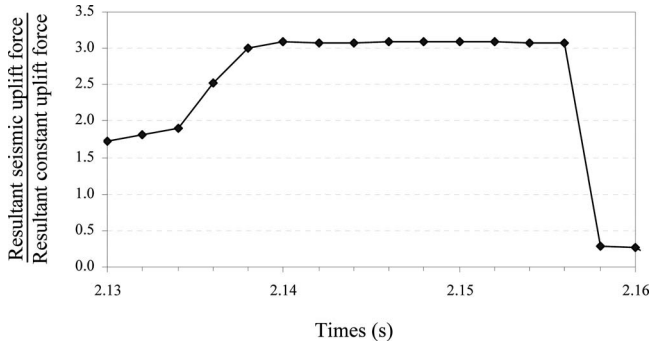

Fig. 15. Resultant uplift force of typical dam at moment of losing contact for peak acceleration 0.5 g, water level 50 m, and residual aperture 75 mm

denly since the majority of the crack becomes void. This indicates that the water flow in the crack behaves like a wedge with a time-force relationship shown in Fig. 15.

Drifting Phenomenon

Once the pivot loses contact because of the water pressure, the whole dam will float in the water. Although the period is very short, the dam will obviously be drifting in a direction depending on the hydro pressure and the ground movement. If the ground has a relatively large acceleration in the downstream direction, the dam will have an upstream drifting as seen in Fig. 11. Nevertheless, the dam will normally experience a downstream drifting driven by the downstream hydro pressure when the downstream ground acceleration is not large enough or it is in the upstream direction.

The drifting phenomenon was noticed in the seismic analysis of cracked gravity dams for constant uplift pressure assumption under strong earthquakes (Pekau and Cui 2004; Pekau and Zhu 2006), as well as in experiments (Tomas et al. 1999), where the uplift of the pivot was the consequence of severe impact. In the present case of seismic uplift pressure, however, this phenomenon occurs prior to the impact, when the upper mouth of the crack is closing. To demonstrate the importance of drifting motion, the drifting components of the overall sliding of the typical dam are listed in Table 2. It is obvious that the higher the water level and the less the residual aperture, the larger the percentage of drifting will be. For instance, with a water level of 61.25 m and residual aperture of 50 mm, the percentages can be as high as 69.01 and 62.91% for peak accelerations of 0.3 and 0.5 g, respectively, whereas there is no drifting for constant uplift pressure in these cases.

Table 2. Accumulated Downstream Drifting (m) of Typical Dam (Numbers in Parentheses Are Percentages of Overall Residual Sliding)

Peak acceleration (g)	Water level (m)	Residual aperture 50 mm	Residual aperture 75 mm	Constant uplift case
0.3	50.00	0.001 (23.73%)	0.000 (0.00%)	0.000
	57.50	0.036 (52.78%)	0.009 (35.65%)	0.000
	61.25	0.382 (69.01%)	0.032 (39.95%)	0.000
0.5	50.00	0.018 (48.26%)	0.007 (33.74%)	0.000
	57.50	0.116 (61.65%)	0.055 (47.60%)	0.000
	61.25	0.435 (62.91%)	0.127 (45.86%)	0.000

Concluding Remarks

A seismic uplift pressure model is developed for penetrated cracks in gravity dams. With assumptions of rigid crack wall and of one-dimensional water flow, the distribution of seismic water pressure can be determined by the openings at both mouths of the crack, as well as by their closing or opening velocities. It is found that the predicted water pressure is consistent with available results.

Computations show that considering the seismic uplift pressure will markedly affect the predicted seismic behavior of the cracked gravity dam. Specifically, the residual sliding for minor earthquakes may be severely underestimated with constant uplift pressure. Both the theoretical and numerical analyses demonstrate that the mechanism of movement is changed. Insignificant rocking, which was usually neglected, may initiate drifting motion because of the seismic uplift pressure, which increases the instability of the cracked dam.

The dynamic water flow plays the role of a wedge to lift the pivot when the cracked dam is rocking. The uplift pressure increases when the upper mouth of the crack is closing until it reaches a certain limit at which the toe of the dam will lose its contact on the crack. Then the seismic pressure remains roughly constant before the dam takes a position inclined to the upstream. Afterwards, the uplift pressure drops abruptly due to the cavitation along the crack.

The residual aperture u_r, which does not explicitly appear in the proposed model, is an important parameter for the application of the proposed model. It is crack dependent and vital in the determination of the seismic uplift pressure when the opened mouth of the crack is at the instance of impact. According to the computations, the lesser the residual aperture, which represents lesser damage while the crack was created, the larger the residual sliding for the concrete gravity dam with a penetrated crack. However, the determination of the residual aperture for a specific crack is left for future study.

This study is based on the assumption of straight and elastic crack surfaces. The aggregate interlock effect is considered through the effective coefficient of friction. When the drifting phenomenon occurs, however, the effects of the aggregate interlock are neglected. It is obvious that the drifting of the dam will be blocked by the asperities along the crack in the form of impact in the horizontal direction, which will cause damage at the asperities such that the effect of the aggregate interlock is weakened. This would in turn increase the overall sliding. This suggests that the drifting component observed in this study may be transformed to additional sliding displacement due to the reduction of the effective coefficient of friction. Therefore, consideration of nonlinear properties of the concrete is needed in future studies.

Acknowledgments

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